

Conditional prob for a 2D multivariate Gaussian

The multivariate Gaussian gives us:

$$\sim p(y_m, y_o) = \frac{1}{\sqrt{(2\pi)^2 \cdot |\Sigma|}} \cdot e^{\left(-\frac{(y-\mu)^T (y-\mu)}{2 \Sigma}\right)}$$

But we want $p(y_m | y_o)$. Considering,

$$\sim \mu = (\mu_m, \mu_o), \quad \Sigma = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \equiv \begin{pmatrix} ()_{mm} & ()_{mo} \\ ()_{om} & ()_{oo} \end{pmatrix}$$

$$\Sigma^{-1} = \frac{1}{ad - b^2} \begin{pmatrix} d & -b \\ -b & a \end{pmatrix}$$

and the exponent becomes:

$$\begin{aligned} \frac{(y-\mu)^T (y-\mu)}{2 \Sigma} &= (y-\mu)^T \frac{1}{2(ad-b^2)} \begin{pmatrix} d & -b \\ -b & a \end{pmatrix} (y-\mu) \\ &= \frac{d \delta_m^2}{2(ad-b^2)} - \frac{2b \delta_m \delta_o}{2(ad-b^2)} + \underbrace{\frac{a \delta_o^2}{2(ad-b^2)}}_{\text{CONSTANT}} \end{aligned}$$

Considering

$$(d \delta_m - b \delta_o, -b \delta_m + a \delta_o) \begin{pmatrix} \delta_m \\ \delta_o \end{pmatrix} = d \delta_m^2 - b \delta_o \delta_m - b \delta_m \delta_o + a \delta_o^2$$

$$\sim p(y_m | y_o) \propto e^{(\text{const})} \cdot e^{\left(-\frac{1}{2v} \cdot (\delta_m - s)^2\right)}, \quad \frac{1}{2v}(\delta_m - s)^2 = \frac{1}{2v}(\delta_m^2 - 2s\delta_m + s^2)$$

$$\sim \frac{d}{2(ad-b^2)} \delta_m^2 - \frac{2b\delta_o}{2(ad-b^2)} \delta_m \xrightarrow{\text{match}} - \underbrace{\frac{1}{2v}}_A \delta_m - \underbrace{\frac{s}{v}}_B \delta_m + \frac{s^2}{v}$$

$$A : \frac{d}{2(ad-b^2)} = \frac{1}{2v} \rightsquigarrow v = a - b^2/d$$

$$B : \frac{b\delta_o}{2(ad-b^2)} = \frac{s}{2v} \rightsquigarrow s = \frac{b}{d} \delta_o$$

$$p(y_m | y_o) \propto e^{(\text{const})} \cdot e^{\left(\frac{1}{2(a-b^2/d)} \left(\delta_m - \frac{b}{a} \delta_o\right)^2\right)}$$